FACULTY OF COMPUTING
UTM Johor Bahru

SECJ 2203: Software Engineering

Semester 01, 2024/2025

PROJECT PROPOSAL

KADA ONLINE SYSTEM

KADA ORGANIZATION

Team Name: Tech Hi-Five

Team Members:

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3. Goe Jie Ying A23CS0224
4. Lam Yoke Yu A23CS0233
5. Teh Ru Qian A23CS0191

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1. Introduction

This proposal aims to propose a system that aims to address the member application, financing application and reporting processes faced by Koperasi Kakitangan KADA Kelantan Berhad, a cooperative under the Lembaga Kemajuan Pertanian Kemubu (hereinafter referred to as "the cooperative") [2].

By using the NABC framework which are the need (N), approach (A), benefit (B) and competitor (C), the Koperasi Kakitangan KADA Kelantan Berhad system can be explained as follows:

Need (N)	The cooperative is currently facing problems related to the manual processes of the coorperative such as the member registration and financing application process. These manual processes are time-consuming, inefficient, and may cause error and data loss. Therefore, it is critical to create a system that helps the coorperative to computerise these manual processes in order to improve the operational efficiency and reduce the worker's burden.
Approach (A)	The proposed solution is to create an online system to computerise the manual processes. The system would digitalise the registration of new members, automate financing applications, enable secure data storage, and generate financial reports. The system will be accessible by multiple devices such as mobile phone and desktop, allowing users to do their registration and application easily without going to the coorperative. Besides, the proposed system provides role-based access for new users, members, board of directors, and admins of the cooperative.
Benefit (B)	By converting the manual processes to an automated system, the coorperative will be benefited in their working efficiency, data accuracy, and security of data. The system will reduce manual error and speed up the process of managing applicants' applications. Moreover, the report generating function would ease the cooperative board of directors to make better decisions for the cooperative.
Competitor (C)	The proposed system has no direct competitors because this proposed system will be the first system used by the cooperative. This system could fulfil the cooperative needs because it was customised based on the functional requirements and design specifications to convert the cooperative manual processes to an automated system.

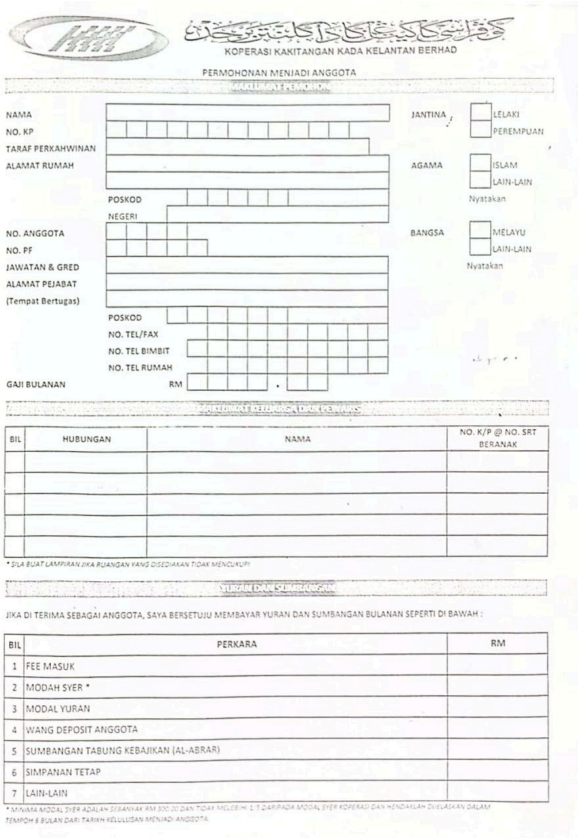
2. Existing Systems

The existing system for managing applications is entirely manual. Applicants are required to fill out and submit their membership and loan applications via physical forms. Staff members must manually verify the information and check the eligibility of each applicant. Once reviewed by the board of directors, applicants are notified of the outcome of their applications. The physical forms used are as follows:

2.1 Member Registration Form

In order to become a member of the cooperative, applicants are required to fill in the member application form (Borang Permohonan Menjadi Anggota). The applicant must be a member of the Lembaga Kemajuan Pertanian Kemubu (KADA) to be eligible to apply for the membership.

Figure 1: Member Application Form



KOPERASI KAKITANGAN KADA KELANTAN BERHAD

PERMOHONAN MENJADI ANGGOTA

NAMA
NO. KP
TARAF PERKAHWINAN
ALAMAT RUMAH
POSKOD
NEGERI
NO. ANGGOTA
NO. PF
JAWATAN & GRED
ALAMAT PEJABAT
(Tempat Bertugas)
POSKOD
NO. TEL/FAX
NO. TEL BIMBIT
NO. TEL RUMAH
GAJI BULANAN RM

JANTINA LELAKI PEREMPUAN
AGAMA ISLAM LAIN-LAIN Nyatakan
BANGSA MELAYU LAIN-LAIN Nyatakan

BIL	HUBUNGAN	NAMA	NO. K/P @ NO. SRT BERANAK

* SILA BUAT LAMPIRAN JIKA RUANGAN YANG DISEDIAKAN TIDAK MENCUKUPI

JIKA DI TERIMA SEBAGAI ANGGOTA, SAYA BERSETUJU MEMBAYAR YURAN DAN SUMBANGAN BULANAN SEPERTI DI BAWAH :

BIL	PERKARA	RM
1	FEE MASUK	
2	MODAH SYER *	
3	MODAL YURAN	
4	WANG DEPOSIT ANGGOTA	
5	SUMBANGAN TABUNG KEBAJIKAN (AL-ABRAR)	
6	SIMPANAN TETAP	
7	LAIN-LAIN	

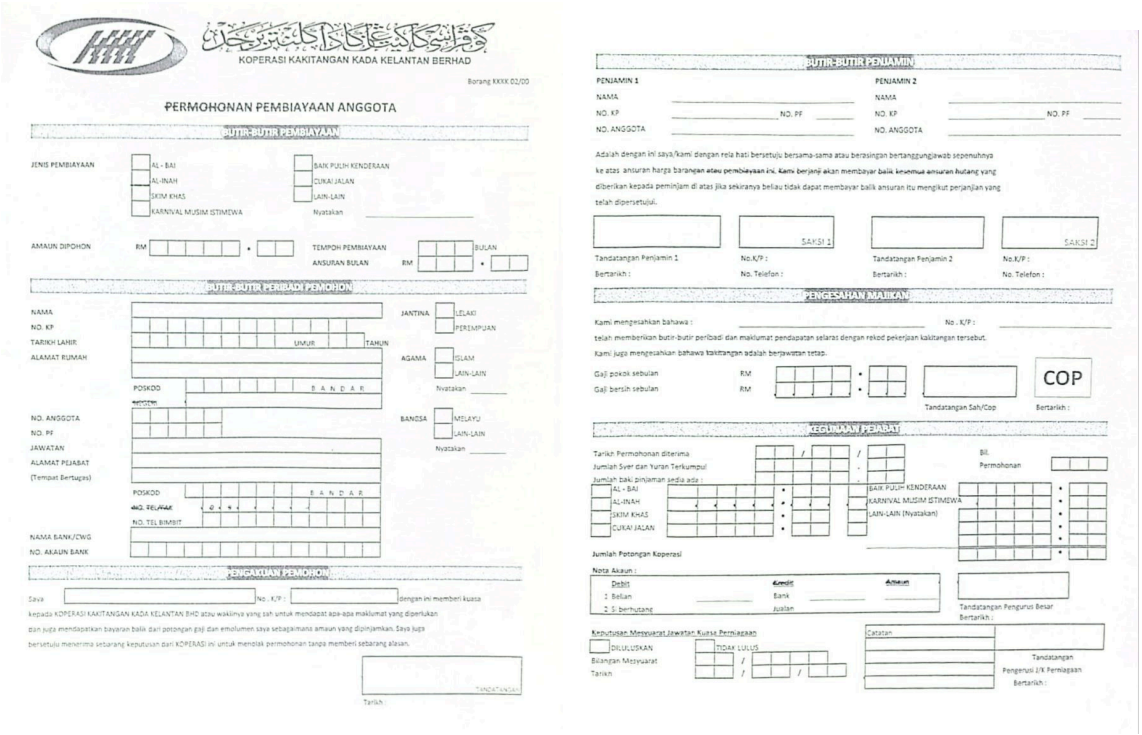
* MINIMA MODAL SYER ADALAH SEBANYAK RM 300.00 DAN TIDAK MELEBIHI 1/5 DARIPADA MODAL SYER KOPERASI DAN HENDAKLAH DILELASKAN DALAM TEMPOH 6 BULAN DARI TARIKH KELULUSAN MENJADI ANGGOTA

2.2 Member Financing Application Form

When a member of the cooperative wishes to apply for financing, they must fill out and submit the member financing application form (Borang Permohonan Pembiayaan Anggota). The applicant's information must be verified by their employer before submission.

Additionally, the applicant must meet the minimum share capital (modah syer) requirement of RM 300.00 to be eligible for financing].

Figure 2: Member Financing Application Form



KOPERASI KAKITANGAN KADA KELANTAN BERHAD

Borang KKKK 02/00

PERMOHONAN PEMBIAYAAN ANGGOTA

BUTIR-BUTIR PEMBIAYAAN

JENIS PEMBIAYAAN: AL - BAI; AL-INAH; SKIM KHAS; KARNIVAL MUSIM ISTIMEWA; BAIK PULIH KENDERAAN; CUKAI JALAN; LAIN-LAIN; Nyatakan

AMAUN DIPOHON RM

TEMPOH PEMBIAYAAN BULAN

ANSURAN BULAN RM

BUTIR-BUTIR PERIBADI PEMOHON

NAMA

NO. KP

TARIKH LAHIR UMUR TAHUN

ALAMAT RUMAH

POSKOD BANDAR

NEGERI

NO. ANGGOTA

NO. PF

JAWATAN

ALAMAT PEJABAT (Tempat Bertugas)

POSKOD BANDAR

NO. PELAPAK

NO. TEL BIMBIT

NAMA BANK/CWG

NO. AKAUN BANK

JANTINA: LELAKI; PEREMPUAN

AGAMA: ISLAM; LAIN-LAIN; Nyatakan

BANGSA: MELAYU; LAIN-LAIN; Nyatakan

PENGAKUAN PEMOHON

Saya No. K/P: dengan ini memberi kuasa kepada KOPERASI KAKITANGAN KADA KELANTAN BHD atau wakilnya yang sah untuk mendapat apa-apa maklumat yang diperlukan dan juga mendapatkan bayaran balik dari potongan gaji dan emolumen saya sebagaimana amaun yang dipinjamkan. Saya juga bersetuju menerima sebarang keputusan dari KOPERASI ini untuk menolak permohonan tanpa memberi sebarang alasan.

TANDATANGAN

Tarikh :

BUTIR-BUTIR PENJAMIN

PENJAMIN 1: NAMA; NO. KP; NO. PF; NO. ANGGOTA

PENJAMIN 2: NAMA; NO. KP; NO. PF; NO. ANGGOTA

Adalah dengan ini saya/kami dengan rela hati bersetuju bersama-sama atau berasingan bertanggungjawab sepenuhnya ke atas ansuran harga barangan atau pembiayaan ini. Kami berjanji akan membayar balik kesemua ansuran hutang yang diberikan kepada peminjam di atas jika sekiranya beliau tidak dapat membayar balik ansuran itu mengikut perjanjian yang telah dipersetujui.

SAKSI 1 SAKSI 2

Tandatangan Penjamin 1 No.K/P : Bertarikh : No. Telefon :

Tandatangan Penjamin 2 No.K/P : Bertarikh : No. Telefon :

PENGESAHAN MAJIKAN

Kami mengesahkan bahawa : No . K/P :

telah memberikan butir-butir peribadi dan maklumat pendapatan selaras dengan rekod pekerjaan kakitangan tersebut.

Kami juga mengesahkan bahawa kakitangan adalah berjawatan tetap.

Gaji pokok sebulan RM

Gaji bersih sebulan RM

COP

Tandatangan Sah/Cop Bertarikh :

TERTIMBAN PEMBAYAR

Tarikh Permohonan diterima

Jumlah Syer dan Yuran Terkumpul

Jumlah baki pinjaman sedia ada :

AL - BAI; AL-INAH; SKIM KHAS; CUKAI JALAN; BAIK PULIH KENDERAAN; KARNIVAL MUSIM ISTIMEWA; LAIN-LAIN (Nyatakan)

Bil. Permohonan

Jumlah Potongan Koperasi

Nota Akaun :

Debit	Kredit	Akaun
1 Belian	Bank	
2 Si berhutang	Jualan	

Tandatangan Pengurus Besar Bertarikh :

Keputusan Mesyuarat Jawatan Kuasa Pembiayaan

DILULUSKAN TIDAK LULUS

Bilangan Mesyuarat

Tarikh

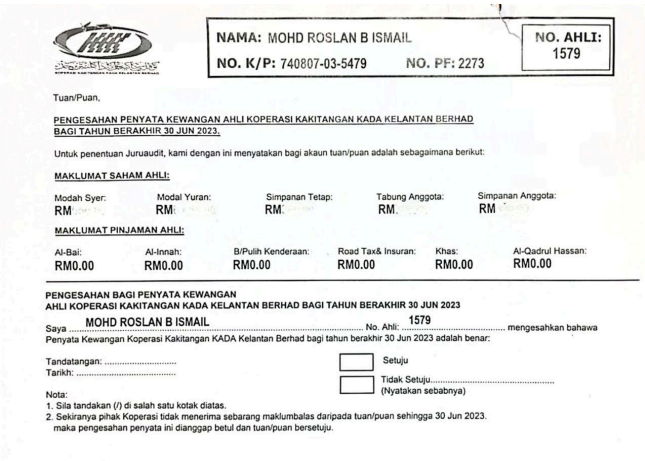
Catatan

Tandatangan Pengerusi J/K Pembiayaan Bertarikh :

2.3 Financial Statement Verification Form

The cooperative also issues a financial statement verification form for members to review the accuracy of their financial status. The financial statement summarises the member's financial status. including sharing information and financing information.

Figure 3: Financial Statement Verification Form



NAMA: MOHD ROSLAN B ISMAIL

NO. K/P: 740807-03-5479 NO. PF: 2273

NO. AHLI: 1579

Tuan/Puan,

PENGESAHAN PENYATA KEWANGAN AHLI KOPERASI KAKITANGAN KADA KELANTAN BERHAD BAGI TAHUN BERAKHIR 30 JUN 2023.

Untuk penentuan Juruaudit, kami dengan ini menyatakan bagi akaun tuan/puan adalah sebagaimana berikut:

MAKLUMAT SAHAM AHLI:

Modah Syer:	Modal Yuran:	Simpanan Tetap:	Tabung Anggota:	Simpanan Anggota:
RM	RM	RM	RM	RM

MAKLUMAT PINJAMAN AHLI:

Al-Bai:	Al-Innah:	B/Pulih Kenderaan:	Road Tax& Insuran:	Khas:	Al-Qadrul Hassan:
RM0.00	RM0.00	RM0.00	RM0.00	RM0.00	RM0.00

PENGESAHAN BAGI PENYATA KEWANGAN
AHLI KOPERASI KAKITANGAN KADA KELANTAN BERHAD BAGI TAHUN BERAKHIR 30 JUN 2023

Saya MOHD ROSLAN B ISMAIL No. Ahli: 1579 mengesahkan bahawa Penyata Kewangan Koperasi Kakitangan KADA Kelantan Berhad bagi tahun berakhir 30 Jun 2023 adalah benar:

Tandatangan: .....
Tarikh: .....

☐ Setuju

☐ Tidak Setuju..... (Nyatakan sebabnya)

Nota:
1. Sila tandakan (/) di salah satu kotak diatas.
2. Sekiranya pihak Koperasi tidak menerima sebarang maklumbalas daripada tuan/puan sehingga 30 Jun 2023. maka pengesahan penyata ini dianggap betul dan tuan/puan bersetuju.

2.4 Comparison between the Existing System and the Proposed System

The proposed system aims to digitise most processes in the existing system. Instead of using physical forms, the system will implement digital forms. This change will make the application process more convenient, as it will not require users to be physically present. Furthermore, a digital system can reduce the risk of data loss, as physical documents are vulnerable to damage, theft, loss, and deterioration over time. Additionally, applicants will have web access to track the status of their applications and view their financial information.

Table 1: Comparison of the existing system and the proposed solution

Features	Existing Manual System	Proposed Digitalise System
Member application	Yes	Yes
Member financing application	Yes	Yes
Track application status	No	Yes
Check loan balance and financial status anytime	No	Yes
Monthly and annually account report of the cooperative	No	Yes
Generation of profit and loss account	No	Yes

3. Proposed System

In the new system, application forms will be shifted online. If the user is not a member, they will need to apply to become a member. If the user is already a member of the cooperative, they will be able to apply for financing via the system. Applicants can apply for membership and financing online without the need for physical presence. A user can only apply for a loan if they are logged in to the system as a member.

The KADA Online System will transform key operations like member registration, loan applications, and financial reporting by replacing manual processes with a digital platform. It addresses inefficiencies through an accessible, secure, and user-friendly solution for all stakeholders, including new users, members, administrators, and board directors. The system ensures smooth multi-device access, role-based controls, and secure data management, ultimately improving operational efficiency and user experience.

User Roles:

- New Users: People who want to join the cooperative can register by filling out an online form. After approval by the board, they will receive an email allowing them to log in and use the system.
- Members: Members can make payments, apply for loans, and track their financial status. They can submit loan applications online, attach required documents, and check the progress of their applications.
- Board Directors: They can approve or reject member and loan applications through a dashboard and view financial reports to assist in decision-making during meetings.
- Administrators: They manage the system by updating member information, reviewing loan applications, and ensuring everything runs smoothly.

The system provides appropriate access to features based on user roles, enhancing efficiency and security.

Additionally, the new system may implement data loss prevention procedures to mitigate the risk of data loss, including data backup procedures and encryption for sensitive information.

In the current system, applications are passed to the board of directors for approval. The proposed system will streamline this process, allowing the administrator to store only digital copies of applications. They can print applications only when a board meeting is scheduled, reducing the risk of unauthorised access to paper-based forms.

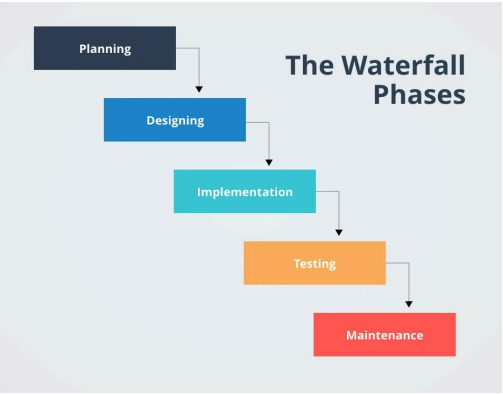
Furthermore, the system can easily generate monthly and annual reports based on the data stored in the database. The reporting mechanism was not mentioned in the current system, so this new feature aims to assist the cooperative in evaluating its condition and financial performance.

The member and loan application processes will be conducted entirely online, eliminating the need for physical forms. Information will be stored in a database, and users will receive automatic email notifications once application results are available. The system will also compute the deduction of monthly salaries and manage members' financial information.

The application approval process will still be manual, involving a board meeting to review applications, after which the results will be entered into the system

4. Software Process Model

For this system, the **waterfall model** is suitable as a software process model because it is easy to manage where it allows for a structured, sequential approach where each phase has specific deliverables before moving to the next. This ensures a well-documented and systematic development process.



4.1 Analysis Phase (Duration: 3 weeks)

In this phase, the main focus is on collecting both functional and non-functional requirements for the system. The process begins with an online meeting session between Encik Ahmad from the cooperative and all members to define the system scope and objectives. During this discussion, we identify the main features of the system, such as member application, loan application and financial report.

Our group leader, Yi Ya leads the requirement gathering process, ensuring that all necessary information is obtained through WhatsApp group discussion with the cooperative. Based on the discussion, we prepare a comprehensive software requirements specification (SRS), which details all functional requirements and non-functional requirements. Jia Lin is responsible for evaluating the current system, identifying weaknesses or errors, and drafting new requirements for the improved system. Additionally, use case diagrams and data flow diagrams (DFD) are created to visually represent system interactions and data processing flows, aiding the stakeholder in understanding the proposed system structure. These deliverables form the basis of the following design phases.

4.2 Design Phase (Duration: 4 weeks)

This phase consists of two main components: High-Level Design (HLD) and Low-Level Design (LLD). HLD provides an overview of the system architecture, outlining the various components, including hardware, software interfaces and user interactions. Jie Ying leads

this phase, ensuring that the design aligns with our business requirements and establishes guidelines for programmers to help them understand the system implementation.

Following the HLD, we move into the LLD phase, which focuses on the specific details of each module within the system. This phase includes defining the logic algorithms, data structures and interfaces required for implementation, serving as a blueprint for developers. At the same time, Yi Ya creates prototypes to enhance customer interactions by Figma, while Yoke Yu is developing a schema to effectively manage data related to users, applications and financial reports. By the end of this, we aim to deliver a set of documentation, including system architecture diagram and UI prototypes as a guideline to proceed into implementation.

4.3 Implementation Phase (Duration: 5 weeks)

This phase begins by setting up the development environment and establishing version control to manage code changes and collaboration effectively. Jie Ying focuses on building the server-side logic for key features such as member registration, loan management, salary deduction and financial reporting. Simultaneously, Ru Qian is responsible for designing and coding the user interface using technologies like HTML, CSS and JavaScript to ensure a seamless user experience.

Once both sides are developed, our teams work together to integrate the front-end with back-end APIs, ensuring that the system's front-end can communicate with the back-end services for functions like displaying member profile management, loan repayment tracking and financial reporting. At the same time, Yoke Yu connects the application to the database to ensure data persistence and managing the storage.

By the end of this phase, the key deliverables include working modules for member registration, loan management, a fully functioning database and successful integration of front-end and back-end systems.

4.4 Testing Phase (Duration: 2 weeks)

It focuses on the correctness and completeness of the system which begins with testing login and main page display for accuracy and proper functionality. Then, our team proceeds with integration testing to make sure that different system components such as loan management and loan repayment tracking are working together seamlessly. After successful integration, system testing is conducted to perform end-to-end testing, simulating real-world scenarios to verify that the system functions as expected under different scenarios.

Additionally, User Acceptance Testing (UAT) is carried out, where a select group of users tests the system and provides feedback on its performance and usability. Ru Qian manages

the UAT process and works closely with stakeholders to ensure that the system meets their expectations.

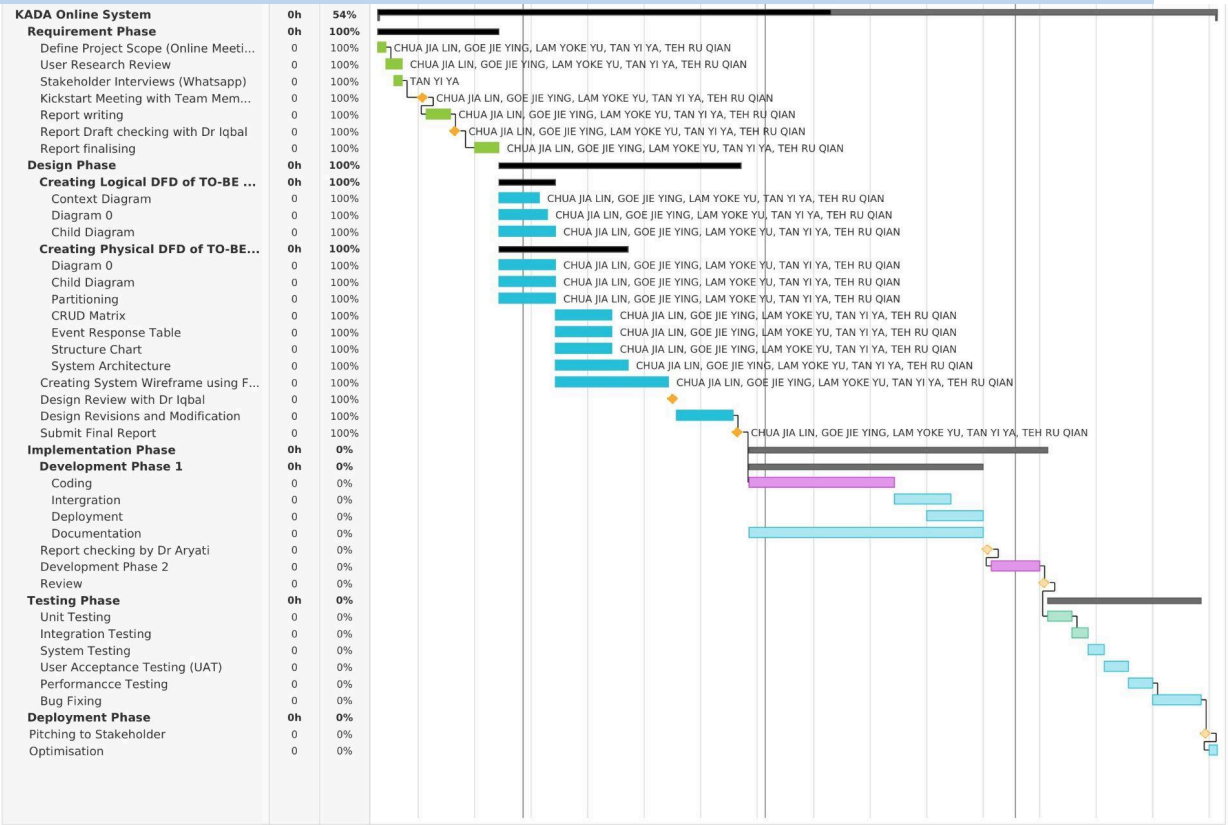
The deliverables from this phase include detailed test reports documenting the testing process and bug reports highlighting any issues found during the testing to help further refine the system.

4.5 Maintenance Phase (Ongoing Post-Deployment)

In the maintenance phase, the focus is on ensuring the system continues to perform optimally after deployment. Ru Qian is responsible for addressing any issues or bugs reported by users to make sure they are resolved promptly to maintain a smooth user experience. Meanwhile, Yoke Yu works on implementing minor updates or improvements based on the user feedback. Therefore, the system remains up-to-date with any changing requirements.

Key deliverables from this phase include detailed bug fix reports,documenting the issues resolved and updated system versions that reflect the latest improvements of the system.

5. Project Schedule



6. References

[1] Sommerville, I. (2016). “Software Engineering”, 10th Edition, US: Pearson.

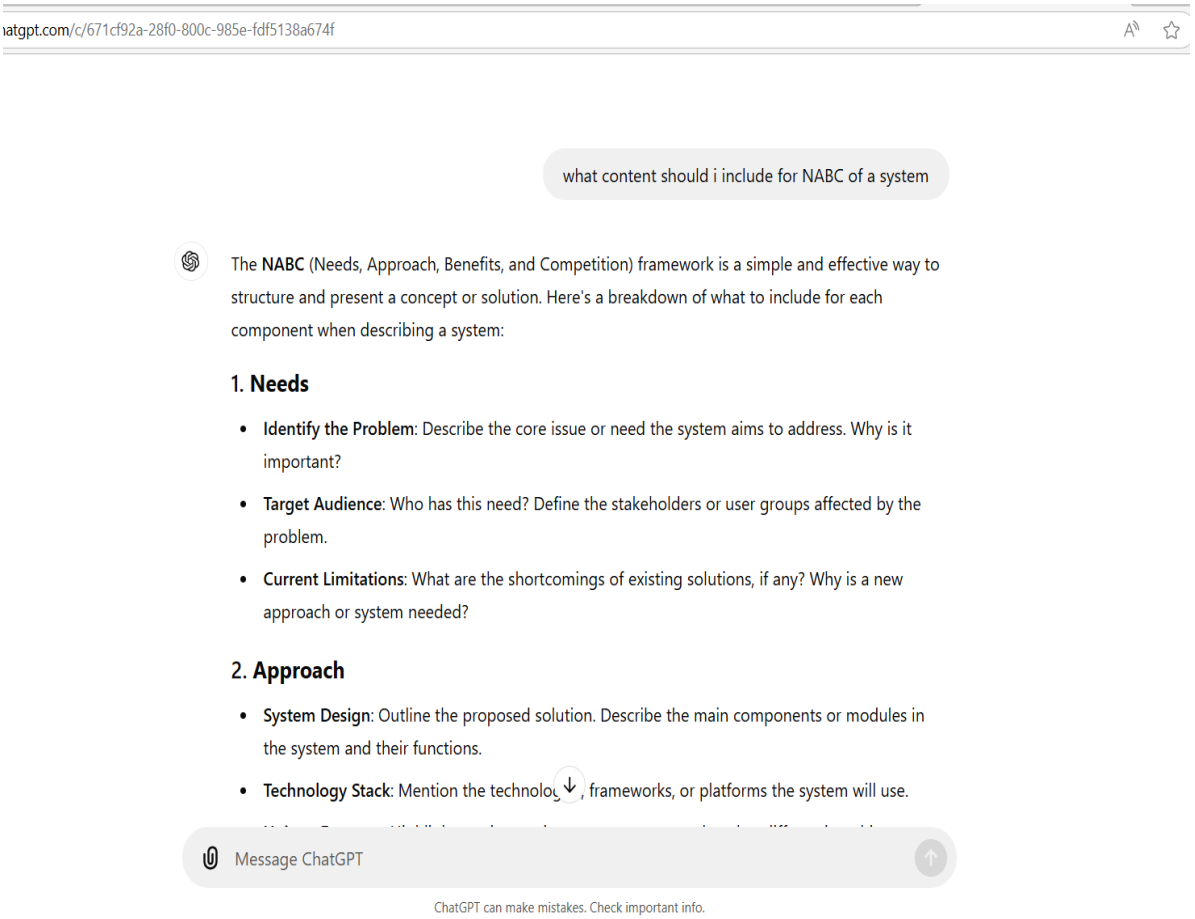
[2] “Portal Lembaga Kemajuan Pertanian Kemubu | KADA KEMUBU.” http://www.kada.gov.my/

[3] R. Wherrity, “Waterfall Project Management: Comprehensive Guide For Success,” Brand, https://www.clearpointstrategy.com/blog/waterfall-project-management (accessed Oct. 19, 2024).

Appendices

For this project proposal, we use ChatGPT to help in completing this proposal.

Screenshots of ChatGPT Conversation:



:hatgpt.com/c/671cf92a-28f0-800c-985e-fdf5138a674f

from traditional methods.

3. Benefits

- **Value Proposition**: Explain how the system addresses the identified needs effectively.
- **Efficiency Gains**: Describe how the system will improve speed, accuracy, cost-effectiveness, or ease of use.
- **User Impact**: Focus on the positive outcomes for the target audience (e.g., increased productivity, reduced workload, improved decision-making).

4. Competition

- **Existing Solutions**: List any current systems or methods that address similar needs.
- **Comparative Advantage**: Detail how your system is better, whether through improved functionality, cost, user experience, or technological advantage.
- **Barriers to Entry**: Mention any obstacles that competitors may face in replicating your approach (e.g., specific expertise, proprietary technology, partnerships).

By focusing on these areas, your NABC will clearly communicate the value and advantage of the system.

Message ChatGPT

ChatGPT can make mistakes. Check important info.

Search

ENG
US